Q6. Show the outputs of the following C code (3 points):

```
#include <stdio.h>
  int y = 30, x = 70, z = 90;
void MyFunction (){
   static int x = 0;
   int y = 0;
   printf("%d,%d,%d \n",x, y, z);
  x++; y++; z++;
}
void main ()
{
   int y = 3, x = 7, z= 9;
  MyFunction ();
  MyFunction ();
  MyFunction ();
}
```

The Output

```
...nteger;

procedure sub1;
begin // {sub1}
  writeln('a = ', a);    ~
  writeln('b = ', b);    30
end; // {sub1}

procedure sub2;
var b : integer;
begin // {sub2}
  a := a + 10;    20
  b := 40;    30        30
  sub1();               40
end; // {sub2}

procedure sub3;
var a : integer;
begin // {sub3}
  a := 20;    20
  b := 30;    30
  sub2();
end; // {sub3}

begin // {main}
  a := 10;
  b := 10;
  sub3();
end. // {main}
```

...output printed under both static and dynamic scope with this call ... calls sub3 *and* sub3 calls sub2 *and* sub2 calls sub1) ?

Static scope	Dynamic scope
20	
0	30

```
...nteger;

procedure sub1;
begin // {sub1}
  writeln('a = ', a);
  writeln('b = ', b);     30
end; // {sub1}

procedure sub2;
var b : integer;
begin // {sub2}
  a := a + 10;   20
  b := 40;   30        30
  sub1();              40
end; // {sub2}

procedure sub3;
var a : integer;
begin // {sub3}
  a := 20;   20
  b := 30;   30
  sub2();
end; // {sub3}

begin // {main}
  a := 10;
  b := 10;
  sub3();
...d. // {main}
```

...output printed under both static and dynamic scope with this call ...n calls sub3 *and* sub3 calls sub2 *and* sub2 calls sub1) ?

Static scope	Dynamic scope
20	30
0	

Q7. Answer the following questions (2 points)

a) What are the possible binding times for a variable? 1.0

b) Do all variables have names? 0.5

NO

c) Explain the difference between name type compatibility and structure type compatibility.

0.25

Name type compatibility

0.25

Structure type compatibility